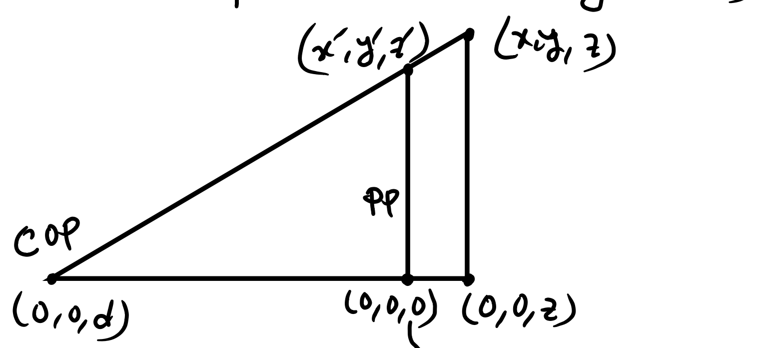
Set A

Q1. (a) Orthographic projection onto the XZ plane at $Y = -7$

(b) Projection plane is at origin $(0,0,0)$



$$\frac{x'}{d} = \frac{x}{d-z} \quad \Big| \quad \frac{y'}{d-z} = \frac{y}{d-z} \quad \Big| \quad z' = 0$$

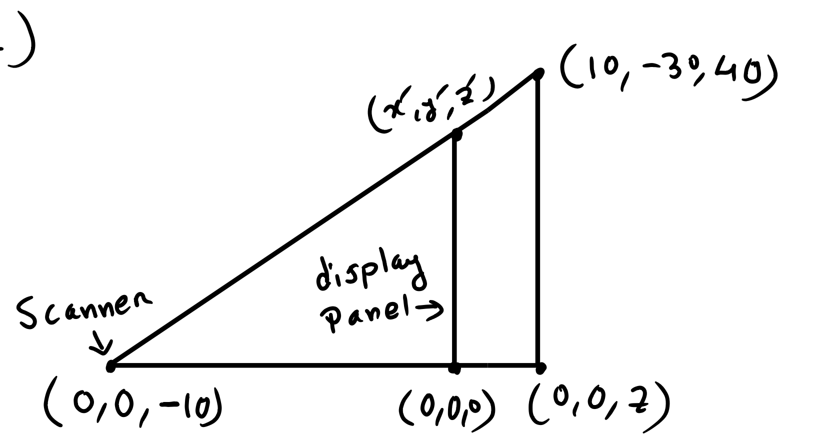
$$\Rightarrow x' = \frac{x}{1-\frac{z}{d}} \quad \Big| \quad \Rightarrow y' = \frac{y}{1-\frac{z}{d}}$$

$$w = 1 - \frac{z}{d}$$

$$\therefore \; x'w = x \qquad y'w = y \qquad z'w = 0 \qquad w = \frac{-1}{d}z + 1$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{-1}{d} & 1 \end{bmatrix}$$

(c)



$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{-1}{-10} & 1 \end{bmatrix} \times \begin{bmatrix} 10 \\ -30 \\ 40 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ -30 \\ 0 \\ 5 \end{bmatrix}$$

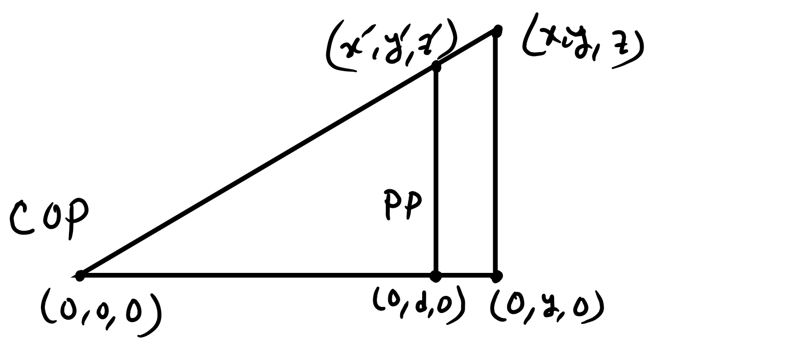
$$\begin{bmatrix} 10 \\ -30 \\ 0 \\ 5 \end{bmatrix} \frac{1}{5} = \begin{bmatrix} 2 \\ -6 \\ 0 \\ 1 \end{bmatrix}$$

$\therefore \; (2, -6, 0)$ (Ans)

Set B

Q1. (a) Orthographic projection onto the XY plane at $Z = -3$

(b) COP is at origin $(0,0,0)$



$$\frac{x'}{d} = \frac{x}{y} \quad \Big| \quad \frac{z'}{d} = \frac{z}{y} \quad \Big| \quad y' = \frac{y}{\left(\frac{y}{d}\right)}$$

$$\Rightarrow x' = \frac{x}{\left(\frac{y}{d}\right)} \quad \Big| \quad \Rightarrow z' = \frac{z}{\left(\frac{y}{d}\right)}$$

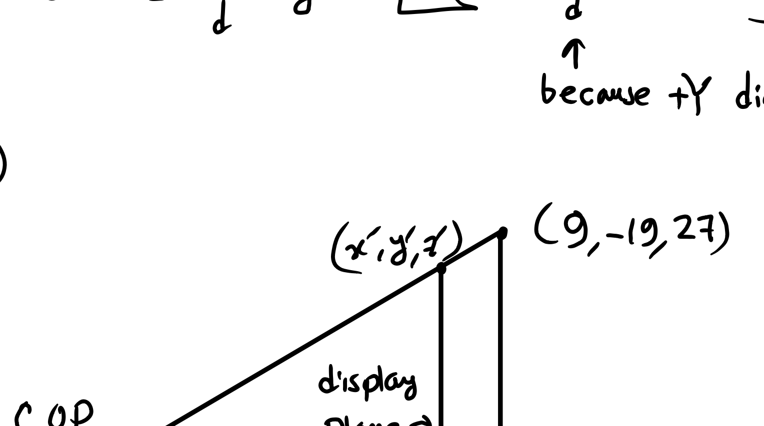
$$W = \frac{y}{d}$$

$$\therefore \; x'w = x \qquad y'w = y \qquad z'w = z \qquad w = \frac{1}{d}y$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & \frac{1}{d} & 0 & 0 \end{bmatrix}$$

↑ because +Y direction

(c)



$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \frac{1}{6} & 0 \end{bmatrix} \times \begin{bmatrix} 9 \\ -19 \\ 27 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ -19 \\ 27 \\ \frac{27}{6} \end{bmatrix}$$

$$\begin{bmatrix} 9 \\ -19 \\ 27 \\ \frac{27}{6} \end{bmatrix} \frac{6}{27} = \begin{bmatrix} 2 \\ -4.22 \\ 6 \\ 1 \end{bmatrix}$$

$\therefore \; (2, -4.22, 6)$ (Ans.)